Creating a SAS Enterprise Miner Project

A SAS Enterprise Miner project contains materials related to a particular analysis task. These materials include analysis process flows, intermediate analysis data sets, and analysis results.

1. Select **File** ⇨ **New** ⇨ **Project** from the main menu. Step 1 of the Create New Project wizard is used to specify the following information:
   - the name of the project you are creating
   - the location of the project

   Type a project name, for example, **Stat223 Project**, in the **Name** field.

   The path specified by the **SAS Server Directory** field is the physical location where the project folder will be created.

2. Select **Next**.

3. To finish defining the project, select **Finish**.

   The SAS Enterprise Miner client application opens the project that you created.

Creating a SAS Library

A SAS library connects SAS Enterprise Miner with the raw data sources, which are the basis of your analysis. A library can link to a directory on the SAS Foundation server, a relational database, or even an Excel workbook.

To define a library, you need to know the name and location of the data structure that you want to link with SAS Enterprise Miner, in addition to any associated options, such as user names and passwords.

Follow the steps below to create a new SAS library.

1. Select **File** ⇨ **New** ⇨ **Library** from the main menu. The Library Wizard – Step 1 of 3 Select Action window appears.

2. Select **Next**. The Library Wizard is updated to show Step 2 of 3 Create or Modify.

3. Type **Stat223** in the **Name** field. Browse to the path where your data are located.

4. Select **Next**. The Library Wizard window is updated to show Step 3 of 3 Confirm Action.

   The Confirm Action window shows the name, type, and path of the created SAS library.

5. Select **Finish**.

   All data available in the SAS library can now be used by SAS Enterprise Miner.

Creating a SAS Enterprise Miner Diagram

A SAS Enterprise Miner diagram workspace contains and displays the steps involved in your analysis.

To define a diagram, you only need to specify its name.

Follow the steps below to create a new SAS Enterprise Miner diagram workspace.

1. Select **File** ⇨ **New** ⇨ **Diagram** from the main menu.

2. Type the name **Predictive Analysis** in the **Diagram Name** field and select **OK**.

   SAS Enterprise Miner creates an analysis workspace window labeled Predictive Analysis

Defining a Data Source

To demonstrate defining a data source, and later, using SAS Enterprise Miner's predictive modeling tools, consider the following specific analysis example:

> A national veterans' organization seeks to better target its solicitations for donation. By only soliciting the most likely donors, less money will be spent on solicitation efforts and more money will be available for charitable concerns. Solicitations involve sending a small gift to an individual and include a request for a donation. Gifts to donors include mailing labels and greeting cards.
>
> The organization has more than 3.5 million individuals in its mailing database. These individuals are classified by their response behaviors to previous solicitation efforts. Of particular interest is the class of individuals identified as *lapsing donors*. These individuals made their most recent donation between 12 and 24 months ago. The organization seeks to rank its lapsing donors based on their responses to a greeting card mailing sent in June of 1997. (The charity calls this the 97NK Campaign.) With this ranking, a decision can be made to either solicit or ignore a lapsing individual in the June 1998 campaign.

The source of this data is the Association for Computing Machinery's (ACM) 1998 KDD-Cup competition. The data set and other details of the competition are publicly available at the UCI KDD Archive at **kdd.ics.uci.edu**.

For model development, the data were sampled to balance the response and non-response rates. (The reason and consequences of this action are discussed in later chapters.) In the original campaign, the response rate was approximately 5%.

The following demonstrations show how to access the 97NK campaign data in SAS Enterprise Miner. The process is divided into three parts:

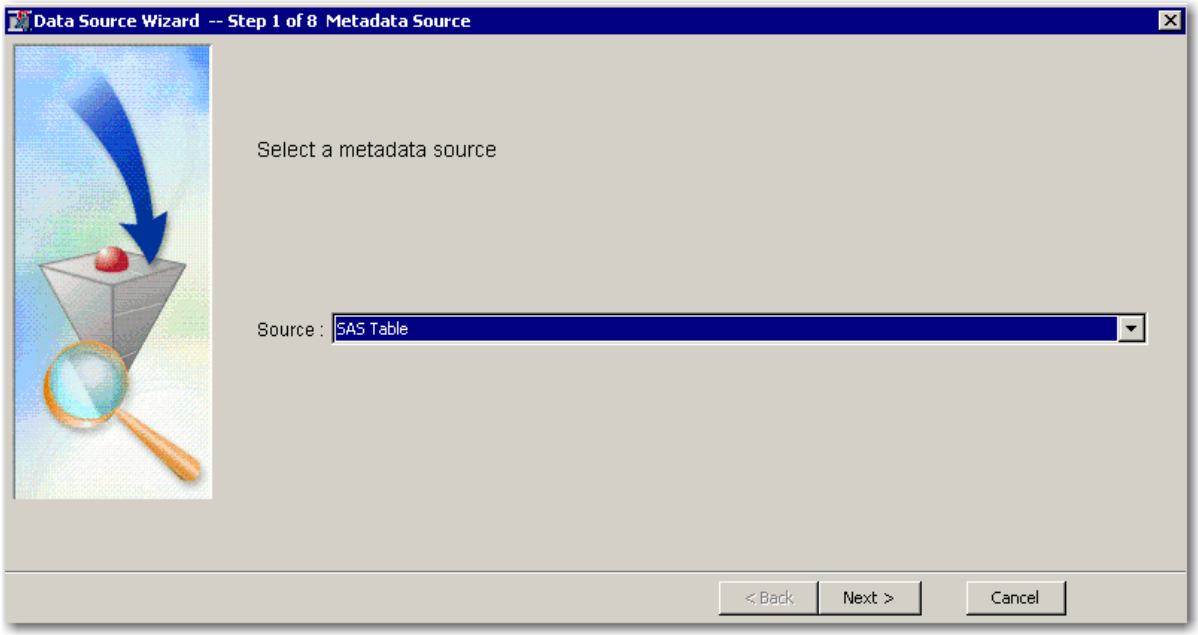
- specifying the source data
- setting the columns metadata
- finalizing the data source specification

Specifying Source Data

A data source links SAS Enterprise Miner to an existing analysis table. To specify a data source, you need to define a SAS library and know the name of the table that you will link to SAS Enterprise Miner.

Follow these steps to specify a data source.

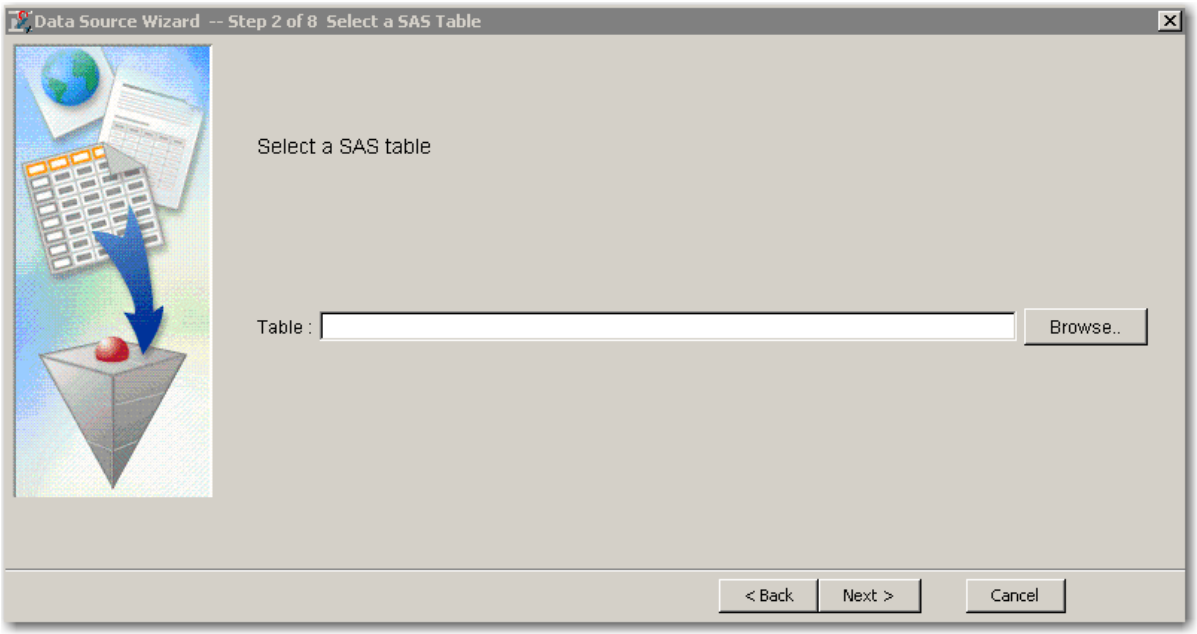
1. Select **File** ⇨ **New** ⇨ **Data Source** from the main menu. The Data Source Wizard – Step 1 of 8 Metadata Source window appears.



The Data Source Wizard guides you through creating a SAS Enterprise Miner data source. Step 1 tells SAS Enterprise Miner where to look for initial metadata values. The default and typical choice is the SAS table that you link to in the next step.

2. Select **Next** to use a SAS table (the common choice) as the source for the metadata.

   The Data Source Wizard continues to Step 2 Select a SAS Table.



   In this step, select the SAS table that you want to make available to SAS Enterprise Miner. You can either type the library name and SAS table name as ***libname.tablename*** or select the SAS table from a list.

3. Select **Browse** to choose a SAS table from the libraries that are visible to the SAS Foundation Server. The Select a SAS Table window appears.

4. Double-click **Stat223**, which is the library name defined in the Library Wizard. The panel on the right is shows the contents of the library.

5. Select the **Pva97nk** SAS table.

6. Select **OK**. The Select a SAS Table window closes and the selected table appears in the **Table** field.

7. Select **Next**. The Data Source Wizard proceeds to Step 3 Table Information.

   This step of the Data Source Wizard provides basic information about the selected table.

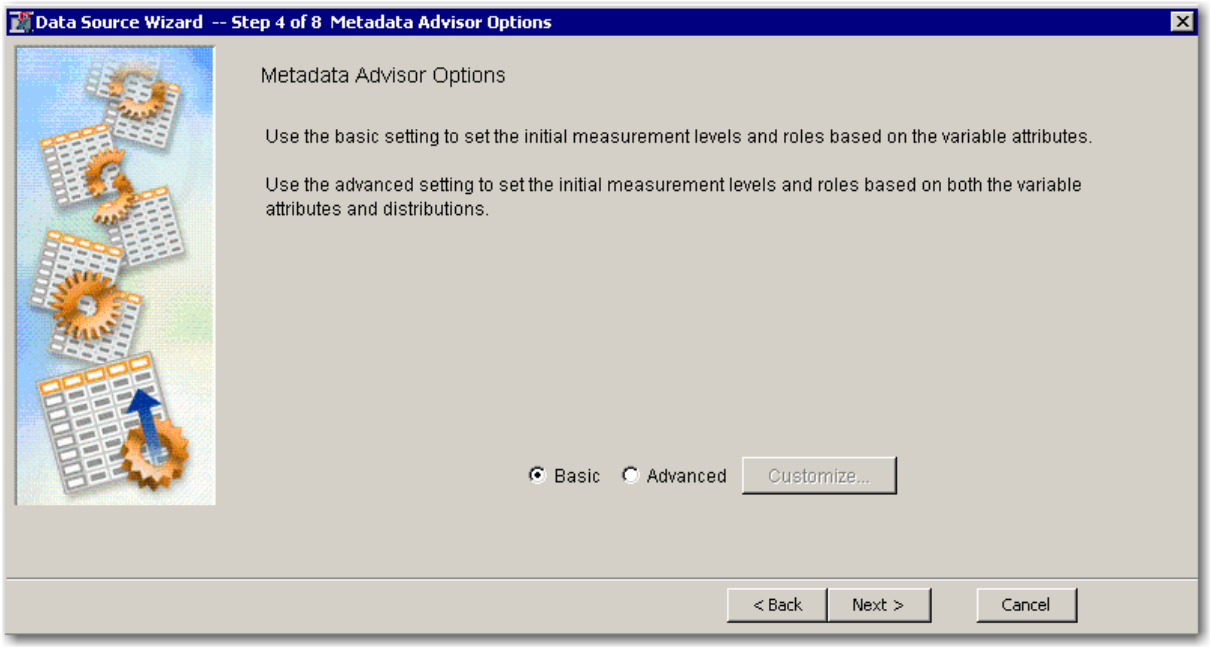
   ✎ The SAS table **PVA97NK** contains 9,686 cases and 28 variables.

Defining Column Metadata

With a data set specified, your next task is to set the column metadata. To do this, you need to know the modeling role and proper measurement level of each variable in the source data set.

Follow these steps to define the column metadata.

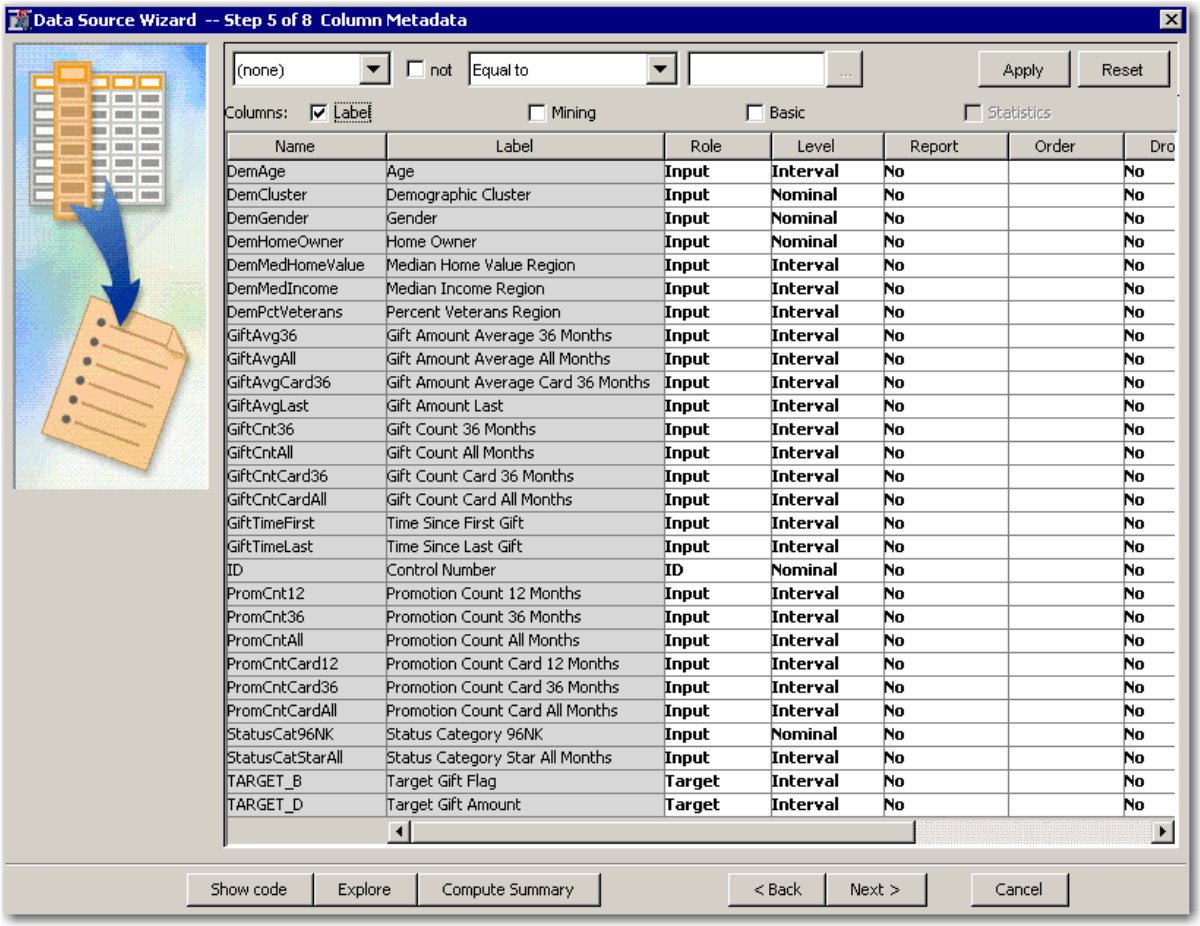
1. Select **Next**. The Data Source Wizard proceeds to Step 4 Metadata Advisor Options.



   This step of the Data Source Wizard starts the metadata definition process. SAS Enterprise Miner assigns initial values to the metadata based on characteristics of the selected SAS table. The Basic setting assigns initial values to the metadata based on variable attributes such as the variable name, data type, and assigned SAS format. The Advanced setting assigns initial values to the metadata in the same way as the Basic setting, but it also assesses the distribution of each variable to better determine the appropriate measurement level.

2. Select **Next** to use the Basic setting.

   The Data Source Wizard proceeds to Step 5 Column Metadata. Select the box next to **Label**.



Name	Label	Role	Level	Report	Order	Dro
DemAge	Age	Input	Interval	No		No
DemCluster	Demographic Cluster	Input	Nominal	No		No
DemGender	Gender	Input	Nominal	No		No
DemHomeOwner	Home Owner	Input	Nominal	No		No
DemMedHomeValue	Median Home Value Region	Input	Interval	No		No
DemMedIncome	Median Income Region	Input	Interval	No		No
DemPctVeterans	Percent Veterans Region	Input	Interval	No		No
GiftAvg36	Gift Amount Average 36 Months	Input	Interval	No		No
GiftAvgAll	Gift Amount Average All Months	Input	Interval	No		No
GiftAvgCard36	Gift Amount Average Card 36 Months	Input	Interval	No		No
GiftAvgLast	Gift Amount Last	Input	Interval	No		No
GiftCnt36	Gift Count 36 Months	Input	Interval	No		No
GiftCntAll	Gift Count All Months	Input	Interval	No		No
GiftCntCard36	Gift Count Card 36 Months	Input	Interval	No		No
GiftCntCardAll	Gift Count Card All Months	Input	Interval	No		No
GiftTimeFirst	Time Since First Gift	Input	Interval	No		No
GiftTimeLast	Time Since Last Gift	Input	Interval	No		No
ID	Control Number	ID	Nominal	No		No
PromCnt12	Promotion Count 12 Months	Input	Interval	No		No
PromCnt36	Promotion Count 36 Months	Input	Interval	No		No
PromCntAll	Promotion Count All Months	Input	Interval	No		No
PromCntCard12	Promotion Count Card 12 Months	Input	Interval	No		No
PromCntCard36	Promotion Count Card 36 Months	Input	Interval	No		No
PromCntCardAll	Promotion Count Card All Months	Input	Interval	No		No
StatusCat96NK	Status Category 96NK	Input	Nominal	No		No
StatusCatStarAll	Status Category Star All Months	Input	Interval	No		No
TARGET_B	Target Gift Flag	Target	Interval	No		No
TARGET_D	Target Gift Amount	Target	Interval	No		No

The Data Source Wizard displays its best guess for the metadata assignments. This guess is based on the name and data type of each variable. The correct values for model role and measurement level are found in the **PVA97NK** metadata table on the next page.

A comparison of the currently assigned metadata to that in the **PVA97NK** metadata table (on the next page) shows several discrepancies. While the assigned modeling roles are mostly correct, the assigned measurement levels for several variables are in error.

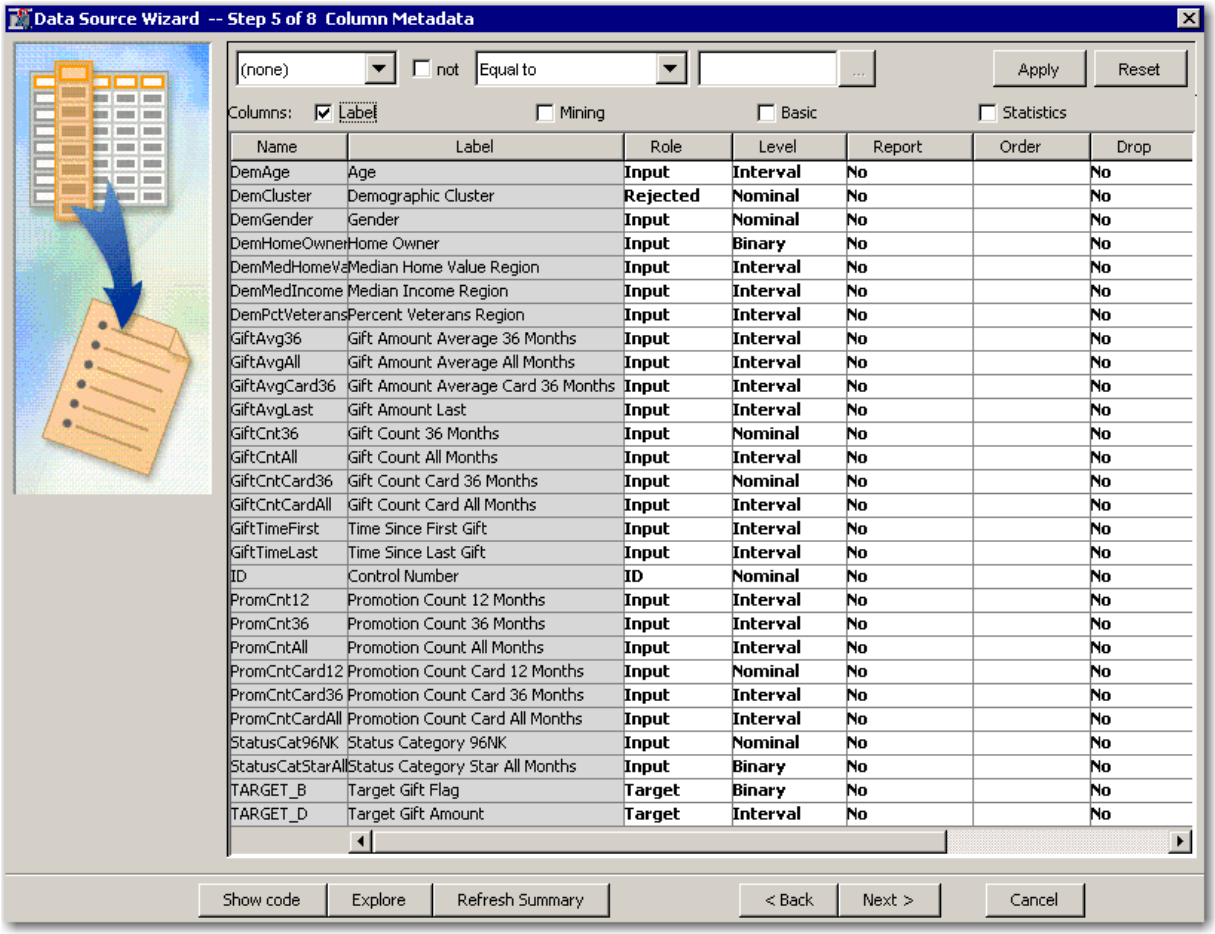
It is possible to improve the default metadata assignments by using the Advanced option in the Metadata Advisor.

3. Select **Back** in the Data Source Wizard. This returns you to Step 4 Metadata Advisor Options.

4. Select the **Advanced** option.

Name	Model Role	Measurement Level	Description
DemAge	Input	Interval	Age
DemCluster	Input	Nominal	Demographic Cluster
DemGender	Input	Nominal	Gender
DemHomeOwner	Input	Binary	Home Owner
DemMedHomeValue	Input	Interval	Median Home Value Region
DemMedIncome	Input	Interval	Median Income Region
DemPctVeterans	Input	Interval	Percent Veterans Region
GiftAvg36	Input	Interval	Gift Amount Average 36 Months
GiftAvgAll	Input	Interval	Gift Amount Average All Months
GiftAvgCard36	Input	Interval	Gift Amount Average Card 36 Months
GiftAvgLast	Input	Interval	Gift Amount Last
GiftCnt36	Input	Interval	Gift Count 36 Months
GiftCntAll	Input	Interval	Gift Count All Months
GiftCntCard36	Input	Interval	Gift Count Card 36 Months
GiftCntCardAll	Input	Interval	Gift Count Card All Months
GiftTimeFirst	Input	Interval	Time Since First Gift
GiftTimeLast	Input	Interval	Time Since Last Gift
ID	ID	Nominal	Control Number
PromCnt12	Input	Interval	Promotion Count 12 Months
PromCnt36	Input	Interval	Promotion Count 36 Months
PromCntAll	Input	Interval	Promotion Count All Months
PromCntCard12	Input	Interval	Promotion Count Card 12 Months
PromCntCard36	Input	Interval	Promotion Count Card 36 Months
PromCntCardAll	Input	Interval	Promotion Count Card All Months
StatusCat96NK	Input	Nominal	Status Category 96NK
StatusCatStarAll	Input	Binary	Status Category Star All Months
TARGET_B	Target	Binary	Target Gift Flag
TARGET_D	Rejected	Interval	Target Gift Amount

5. Select **Next** to use the Advanced setting. The Data Source Wizard again proceeds to Step 5 Column Metadata.



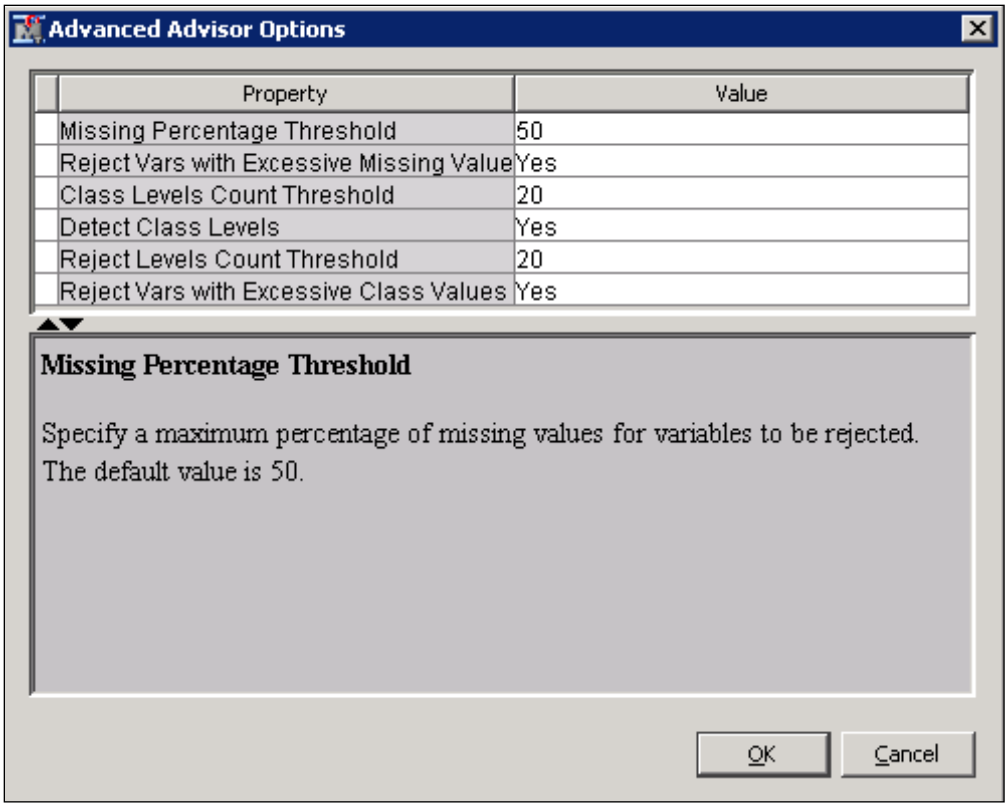
Name	Label	Role	Level	Report	Order	Drop
DemAge	Age	Input	Interval	No		No
DemCluster	Demographic Cluster	Rejected	Nominal	No		No
DemGender	Gender	Input	Nominal	No		No
DemHomeOwner	Home Owner	Input	Binary	No		No
DemMedHomeVa	Median Home Value Region	Input	Interval	No		No
DemMedIncome	Median Income Region	Input	Interval	No		No
DemPctVeterans	Percent Veterans Region	Input	Interval	No		No
GiftAvg36	Gift Amount Average 36 Months	Input	Interval	No		No
GiftAvgAll	Gift Amount Average All Months	Input	Interval	No		No
GiftAvgCard36	Gift Amount Average Card 36 Months	Input	Interval	No		No
GiftAvgLast	Gift Amount Last	Input	Interval	No		No
GiftCnt36	Gift Count 36 Months	Input	Nominal	No		No
GiftCntAll	Gift Count All Months	Input	Interval	No		No
GiftCntCard36	Gift Count Card 36 Months	Input	Nominal	No		No
GiftCntCardAll	Gift Count Card All Months	Input	Interval	No		No
GiftTimeFirst	Time Since First Gift	Input	Interval	No		No
GiftTimeLast	Time Since Last Gift	Input	Interval	No		No
ID	Control Number	ID	Nominal	No		No
PromCnt12	Promotion Count 12 Months	Input	Interval	No		No
PromCnt36	Promotion Count 36 Months	Input	Interval	No		No
PromCntAll	Promotion Count All Months	Input	Interval	No		No
PromCntCard12	Promotion Count Card 12 Months	Input	Nominal	No		No
PromCntCard36	Promotion Count Card 36 Months	Input	Interval	No		No
PromCntCardAll	Promotion Count Card All Months	Input	Interval	No		No
StatusCat96NK	Status Category 96NK	Input	Nominal	No		No
StatusCatStarAll	Status Category Star All Months	Input	Binary	No		No
TARGET_B	Target Gift Flag	Target	Binary	No		No
TARGET_D	Target Gift Amount	Target	Interval	No		No

While many of the default metadata settings are correct, there are several items that need to be changed. For example, the **DemCluster** variable is rejected (for having too many distinct values), and several numeric inputs have their measurement level set to Nominal instead of Interval (for having too few distinct values).

To avoid the time-consuming task of making metadata adjustments, go back to the previous Data Source Wizard step and customize the Metadata Advisor.

6. Select **Back**. You return to the Metadata Advisor Options window.

7. Select **Customize**. The Advanced Advisor Options dialog box appears.



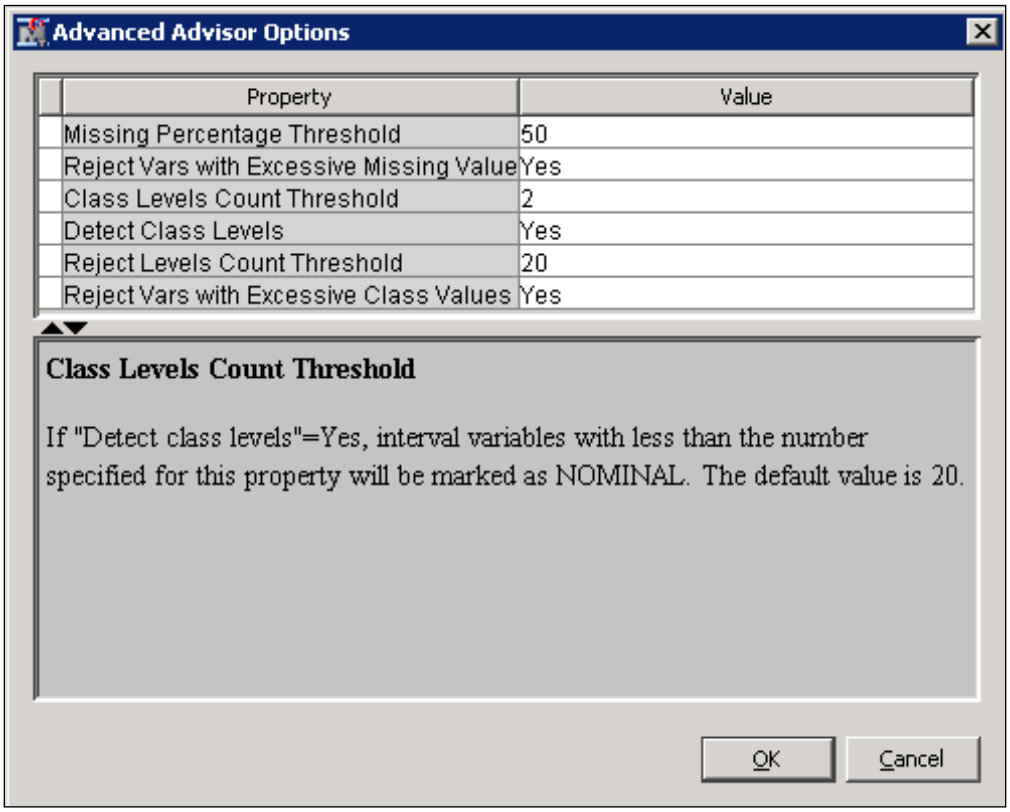
Using the default Advanced options, the Metadata Advisor can do the following:

- reject variables with an excessive number of missing values (default=50%)
- detect the number class levels of ***numeric*** variables and assign a role of Nominal to those with class counts below the selected threshold (default=20)
- detect the number class levels of ***character*** variables and assign a role of Rejected to those with class counts above the selected threshold (default=20)

In the **PVA97NK** table, there are several numeric variables with fewer than 20 distinct values that should ***not*** be treated as nominal. Similarly, there is one class variable with more than 20 levels that should ***not*** be rejected.

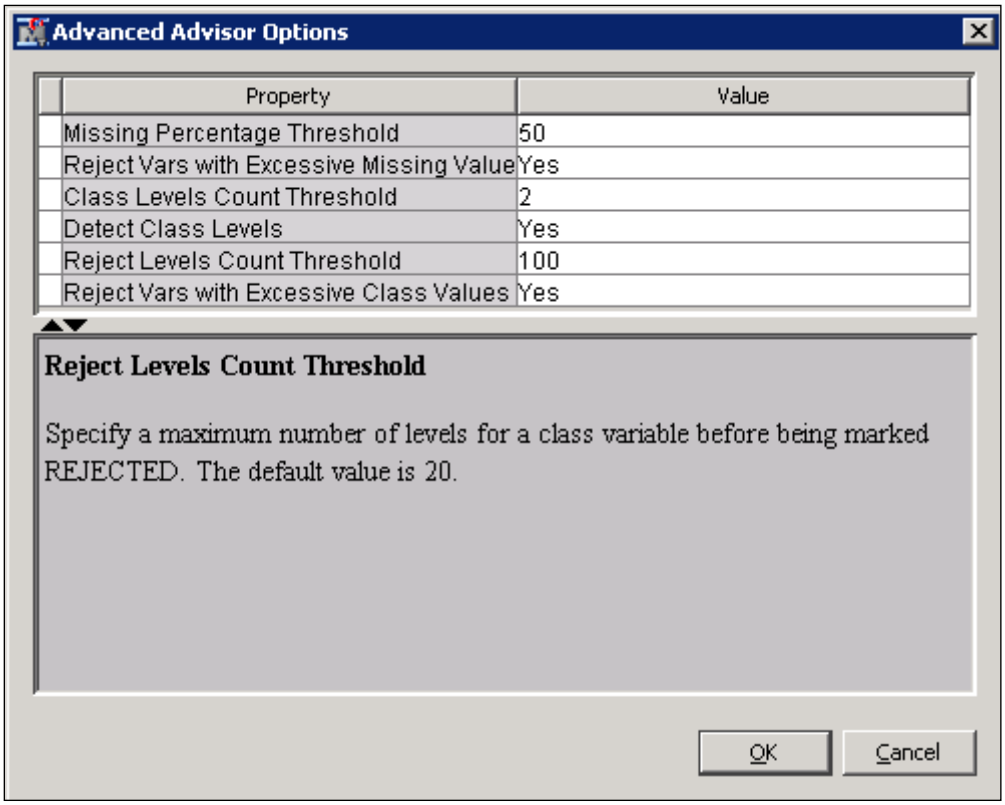
To avoid changing many metadata values in the next step of the Data Source Wizard, you should alter these defaults.

8. Type **2** as the Class Levels Count Threshold value so that only binary numeric variables are treated as categorical variables.



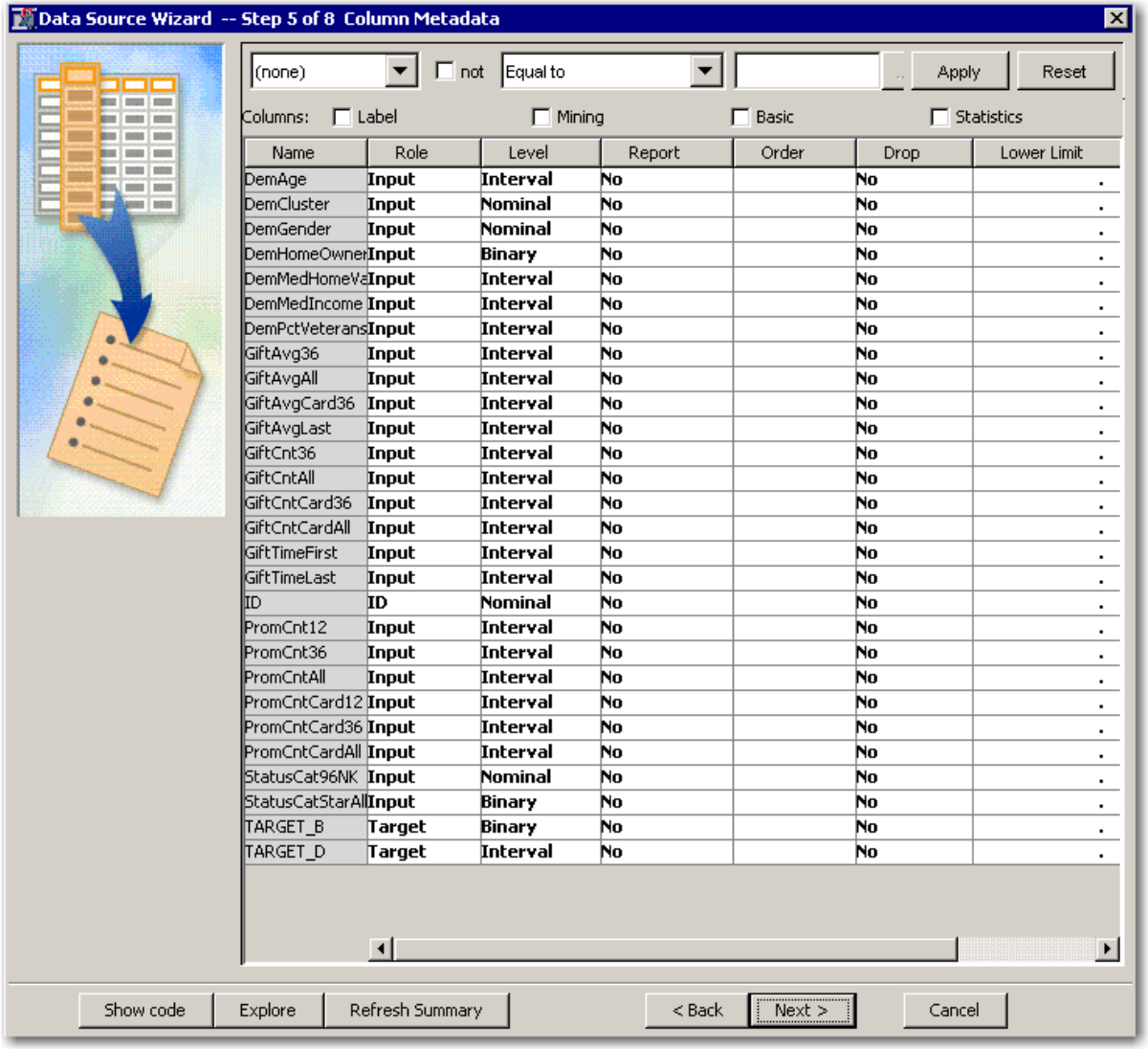
9. Type **100** as the Reject Levels Count Threshold value so that only character variables with more than 100 distinct values are rejected.

Be sure to press ENTER after you type the number **100**. Otherwise, the value might not be registered in the field.



10. Select **OK** to close the Advanced Advisor Options dialog box.

11. Select **Next** to proceed to Step 5 of the Data Source Wizard.

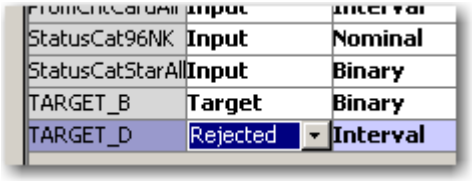


Name	Role	Level	Report	Order	Drop	Lower Limit
DemAge	Input	Interval	No		No	.
DemCluster	Input	Nominal	No		No	.
DemGender	Input	Nominal	No		No	.
DemHomeOwner	Input	Binary	No		No	.
DemMedHomeVa	Input	Interval	No		No	.
DemMedIncome	Input	Interval	No		No	.
DemPctVeterans	Input	Interval	No		No	.
GiftAvg36	Input	Interval	No		No	.
GiftAvgAll	Input	Interval	No		No	.
GiftAvgCard36	Input	Interval	No		No	.
GiftAvgLast	Input	Interval	No		No	.
GiftCnt36	Input	Interval	No		No	.
GiftCntAll	Input	Interval	No		No	.
GiftCntCard36	Input	Interval	No		No	.
GiftCntCardAll	Input	Interval	No		No	.
GiftTimeFirst	Input	Interval	No		No	.
GiftTimeLast	Input	Interval	No		No	.
ID	ID	Nominal	No		No	.
PromCnt12	Input	Interval	No		No	.
PromCnt36	Input	Interval	No		No	.
PromCntAll	Input	Interval	No		No	.
PromCntCard12	Input	Interval	No		No	.
PromCntCard36	Input	Interval	No		No	.
PromCntCardAll	Input	Interval	No		No	.
StatusCat96NK	Input	Nominal	No		No	.
StatusCatStarAll	Input	Binary	No		No	.
TARGET_B	Target	Binary	No		No	.
TARGET_D	Target	Interval	No		No	.

A comparison of the Column Metadata table to the table at the beginning of the demonstration shows that most of the metadata is correctly defined. SAS Enterprise Miner correctly inferred the model roles for the non-input variables by their names. The measurement levels are correctly defined by using the Advanced Metadata Advisor.

The analysis of the **PVA97NK** data in this course focuses on the **TARGET_B** variable, so the **TARGET_D** variable should be rejected.

12. Select **Role** ⇨ **Rejected** for **TARGET_D**.



StatusCat96NK	Input	Nominal
StatusCatStarAll	Input	Binary
TARGET_B	Target	Binary
TARGET_D	Rejected	Interval

In summary, Step 5 Column Metadata is usually the most time-consuming of the Data Source Wizard steps. You can use the following tips to reduce the amount of time required to define metadata for SAS Enterprise Miner predictive modeling data sets:

- Include only variables that you intend to use in the modeling process in your raw data source.
- For variables that are not inputs, use variable names that start with the intended role. For example, an ID variable should start with **ID** and a target variable should start with **Target**.
- Inputs that are to have a nominal measurement level should have a ***character*** data type.
- Inputs that are to be interval must have a ***numeric*** data type.
- Customize the Metadata Advisor to have a Class Level Count set equal to **2** and a Reject Levels Count set equal to a number greater than the maximum cardinality (level count) of your nominal inputs.

Finalizing the Data Source Specification

Follow these steps to complete the data source specification process.

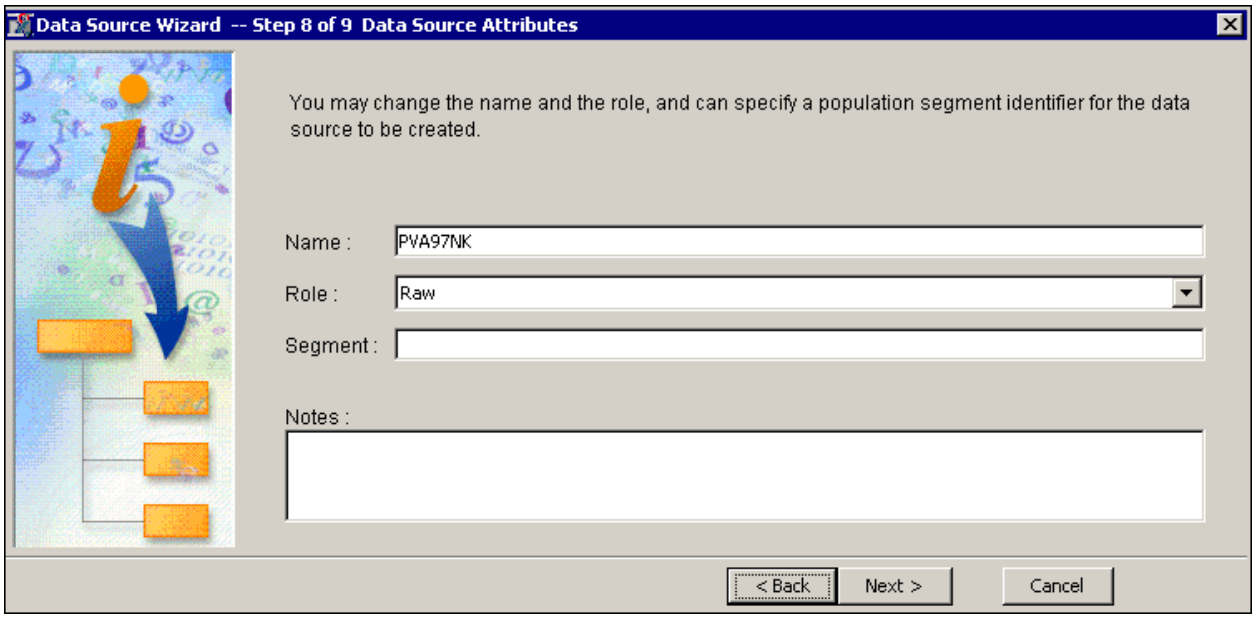
1. Select **Next** to proceed to Decision Configuration.

   The Data Source Wizard gained an extra step due to the presence of a categorical (binary, ordinal, or nominal) target variable.

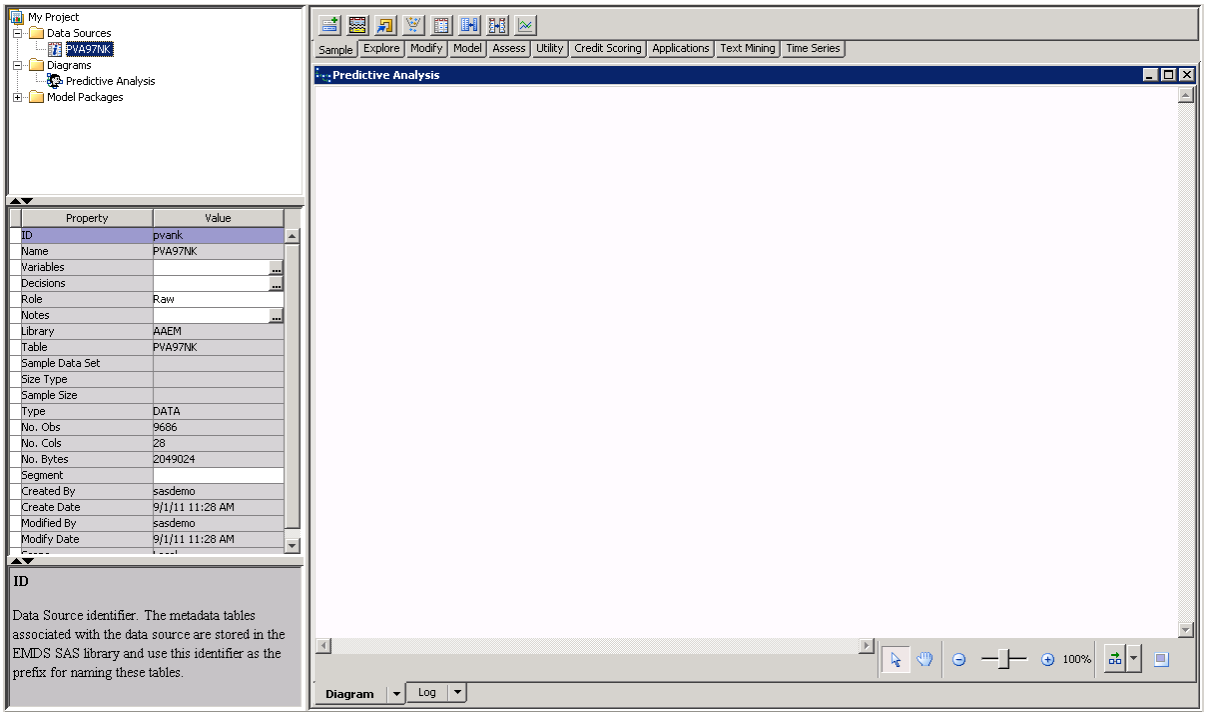
   When you define a predictive modeling data set, it is important to properly configure decision processing. In fact, obtaining meaningful models often requires using these options. The **PVA97NK** table was structured so that reasonable models are produced ***without*** specifying decision processing.

2. Select **Next** to proceed to Create Sample. No sample data set is created, so leave this step set to the default **No.**

3. Select **Next** and you reach the second to the last step of the Data Source Wizard.



This step enables you to set a role for the data source and add descriptive comments about the data source definition. For the upcoming analysis, a table role of Raw is acceptable.

4. Select **Next**. The final step in the Data Source Wizard provides summary details about the data table that you created. Select **Finish**.

The **PVA97NK** data source is added to the Data Sources entry in the Project panel.